TUGAS 2

Pembelajaran Mesin

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Kelas : Pembelajaran Mesin

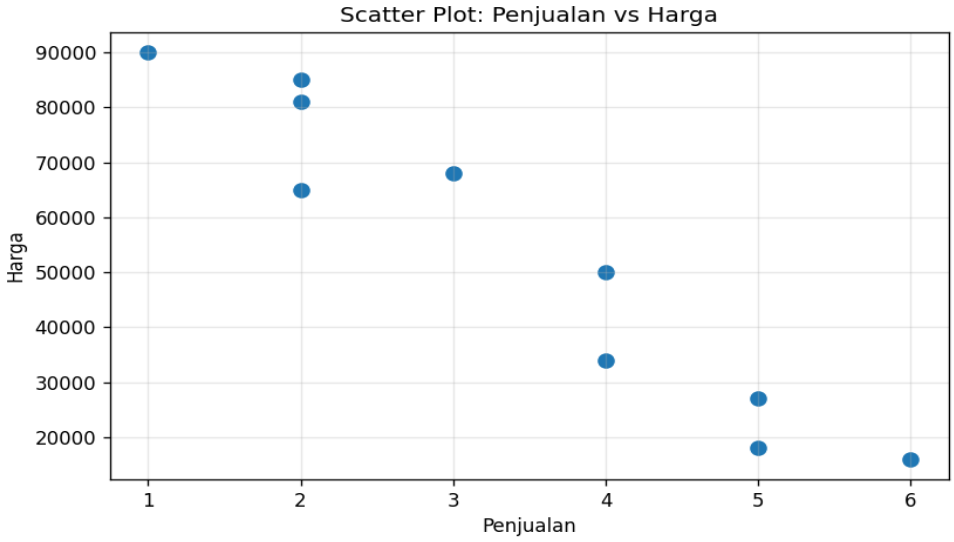
1. Regresi Linear Sederhana – Data Penjualan & Harga

```
#%%
# Import library dan package yang dibutuhkan
import numpy as np # untuk perhitungan saintifik
import matplotlib.pyplot as plt # untuk plotting
from sklearn.linear_model import LinearRegression # import library LinearRegression dari
scikit-learn

#%%
# Membuat data sembarang
penjualan = np.array([6,5,5,4,4,3,2,2,2,1])
harga = np.array([16000, 18000, 27000, 34000, 50000, 68000, 65000, 81000, 85000, 90000])

# %%
# Print data sembarang
print("Data Penjualan :", penjualan)
print("Data Harga :", harga)

# %%
plt.scatter(penjualan, harga)

#%%
# Membuat model regresi
penjualan = penjualan.reshape(-1,1) # tukar baris dan kolom variabel
linreg = LinearRegression()
linreg.fit(penjualan, harga)

#%%
# Visualisasi plot hasil regresi
plt.scatter(penjualan, harga, color='red')
plt.plot(penjualan, linreg.predict(penjualan))
plt.title("Visualisasi model regresi data penjualan dan harga")
plt.xlabel("Penjualan")
plt.ylabel("Harga")
```

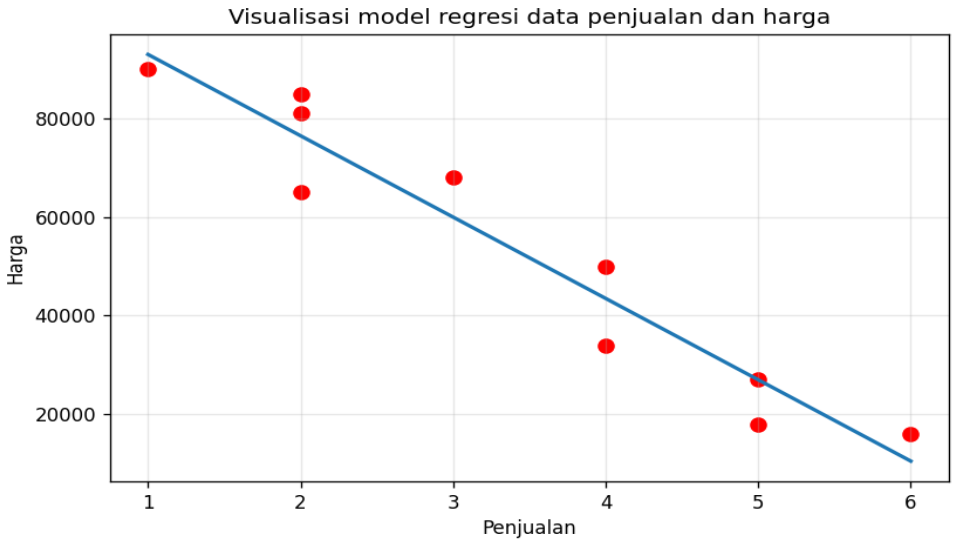
Output – Print Data:

```
Data Penjualan : [6 5 5 4 4 3 2 2 2 1]
Data Harga : [16000 18000 27000 34000 50000 68000 65000 81000 85000 90000]
```

Output – Scatter Plot:



Output – Visualisasi Model Regresi Linear:



Output – Koefisien Model:

```
Coefficients: [-16500.]
Intercept: 109500.0
```

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2. Regresi Linear – Dataset FuelConsumptionCo2

Dataset diambil dari: https://s3-api.us-geo.objectstorage.softlayer.net/cf-courses-data/CognitiveClass/ML0101ENv3/labs/FuelConsumptionCo2.csv

```
# import library dan package yang dibutuhkan
import pandas as pd # untuk dataframe
import matplotlib.pyplot as plt
import numpy as np
from sklearn.linear_model import LinearRegression # import library LinearRegression dari
scikit-learn
```

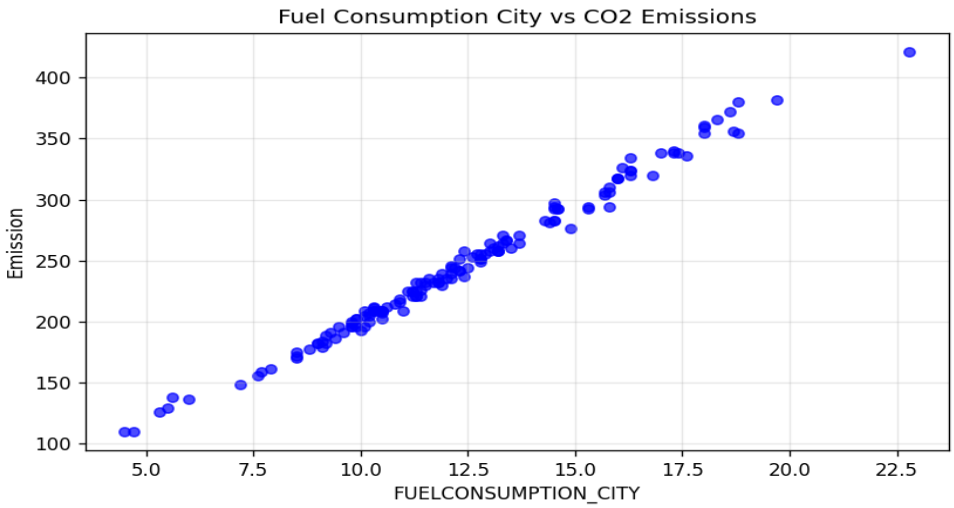
```
#%%
# Membaca data CSV
df = pd.read_csv("FuelConsumptionCo2.csv") # membaca data
# melihat 5 baris pertama data
df.head()

#%%
# kita ambil kolom mana saja yang akan kita analisis, dan membuang sisanya
cdf = df[['ENGINESIZE','CYLINDERS','FUELCONSUMPTION_CITY','CO2EMISSIONS']]
cdf.head(9)

# %%
#%%
# Kita plot hubungannya (Fuel Consumption vs Emission)
plt.scatter(cdf.FUELCONSUMPTION_CITY, cdf.CO2EMISSIONS, color='blue')
plt.xlabel("FUELCONSUMPTION_CITY")
plt.ylabel("Emission")
plt.show()

#%%
# Kita plot hubungannya (Engine Size vs Emission)
plt.scatter(cdf.ENGINESIZE, cdf.CO2EMISSIONS, color='blue')
plt.xlabel("Engine size")
plt.ylabel("Emission")
plt.show()

# %%
#%%
# Membagi data
msk = np.random.rand(len(df)) < 0.8
train = cdf[msk]
test = cdf[~msk]

#%%
# Visualisasi data training antara engine size dan emission
plt.scatter(train.ENGINESIZE, train.CO2EMISSIONS, color='blue')
plt.xlabel("Engine size")
plt.ylabel("Emission")
plt.show()

#%%
# Membuat model regresi
regr = LinearRegression()
train_x = np.asanyarray(train[['ENGINESIZE']])
train_y = np.asanyarray(train[['CO2EMISSIONS']])
regr.fit(train_x, train_y)
# Koefisien model
print ('Coefficients: ', regr.coef_)
print ('Intercept: ', regr.intercept_)

# %%
#%%
# Plot hasil regresi
plt.scatter(train.ENGINESIZE, train.CO2EMISSIONS, color='blue')
```

```
plt.plot(train_x, regr.coef_[0][0]*train_x + regr.intercept_[0], '-r')
plt.xlabel("Engine size")
plt.ylabel("Emission")
```

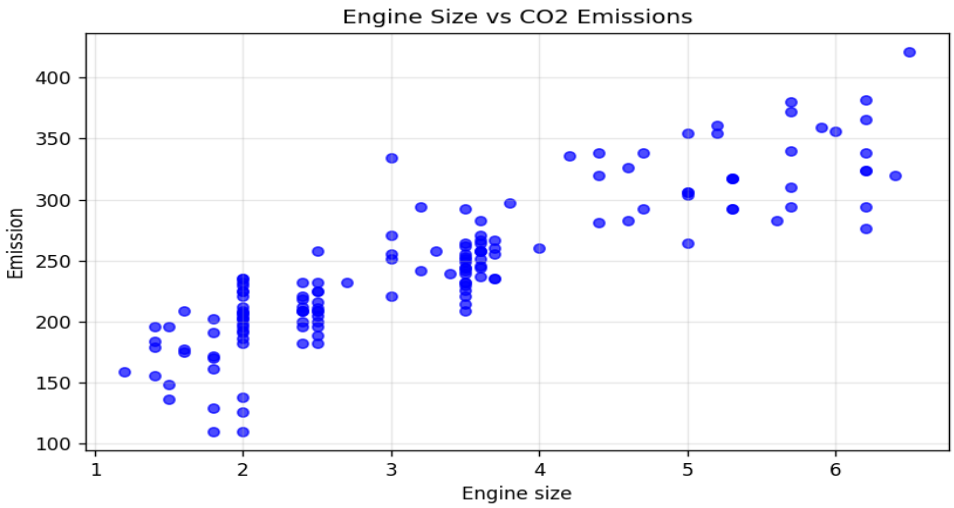
Output – cdf.head(9):

```
ENGINESIZE CYLINDERS FUELCONSUMPTION_CITY CO2EMISSIONS
0 2.0 4 9.9 196
1 2.4 4 11.2 221
2 1.5 4 6.0 136
3 3.5 6 12.7 255
4 3.5 6 12.1 244
5 3.5 6 11.9 230
6 3.5 6 11.8 232
7 3.7 6 12.8 255
8 3.7 6 13.4 267
```

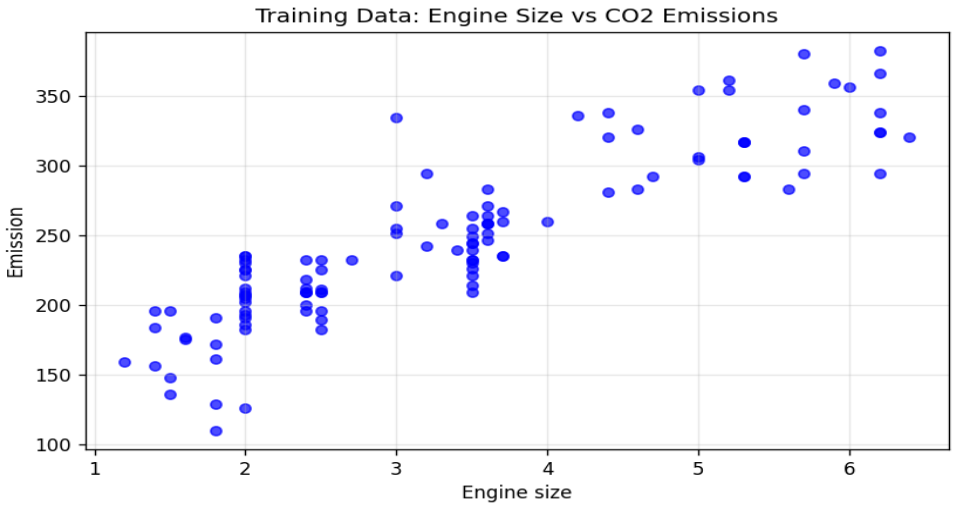
Output – Plot: Fuel Consumption City vs Emission:



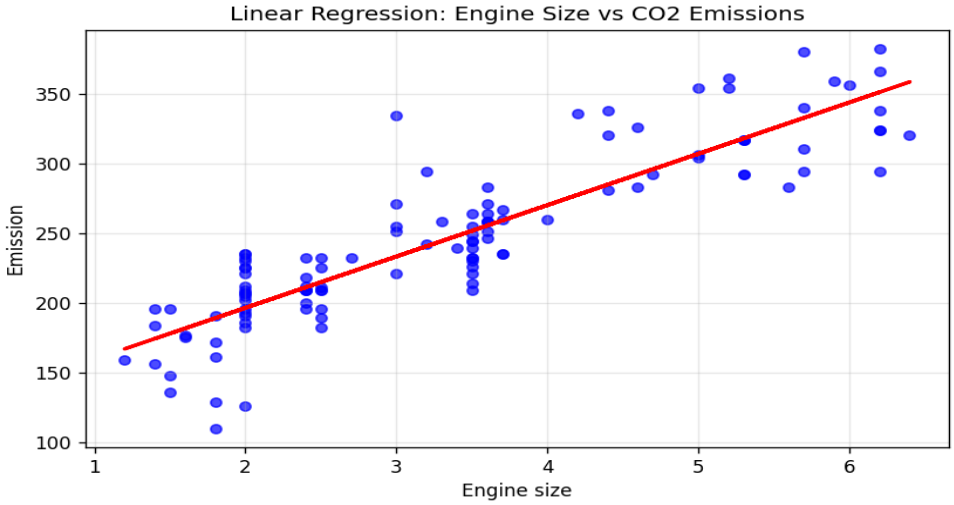
Output – Plot: Engine Size vs Emission:



Output – Visualisasi Data Training:



Output – Koefisien Model Regresi:

```
Coefficients: [[36.8119]]
Intercept: [122.7699]
```

Output – Plot Hasil Regresi: